The sharp Berry–Esseen bound survives small independent noise

28 October 2026

If the two conditional noise laws are identical, the marginal is a Bernoulli variable plus independent noise. Conditioning on that noise allows us to apply the sharp binomial theorem at a random threshold. A finite Taylor replacement then compares the smoothed Gaussian term with the Gaussian of the correct total variance. The increase in the third absolute moment pays for this error. The resulting argument covers every sample size and threshold in this subclass.

Theorem 1. *Let $B \sim \mathrm{Bern}(p)$, where $1/4 \le p \le 3/4$, and let U be independent of B, centered, and supported on an interval of length at most $1/5$. Put*

$$v = p(1-p), \quad s = \mathbb{E}U^2, \quad X = \frac{B - p + U}{\sqrt{v+s}}, \quad \rho = \mathbb{E}|X|^3, \qquad C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}}.$$

For independent copies $X_1, \dots, X_n$, every positive integer n satisfies

$$\sqrt{n}\,\sup_{H\in\mathbb{R}} \left|\mathbb{P}(X_1 + \cdots + X_n \le H) - \Phi(H/\sqrt{n})\right| \le C_{\mathrm{E}}\rho - \frac{3s}{1000 v^{3/2}}. \tag{1}$$

In particular the sharp Berry–Esseen bound holds for this class, with a strict relative-noise margin when $s > 0$. No upper bound on ns, lower bound on n, threshold restriction, acquisition or contact hypothesis is required.

1 The two estimates

The only external probability theorem used is Schulz's sharp binomial bound [1, Theorem 1, printed p. 1]. For $B_n \sim \mathrm{Bin}(n,p)$, $0 < p < 1$ and every $n \ge 1$, it states

$$\sup_{z\in\mathbb{R}} \left|\mathbb{P}(B_n \le z) - \Phi\left(\frac{z-np}{\sqrt{nv}}\right)\right| < \frac{C_{\mathrm{E}}\rho_B}{\sqrt{n}}, \qquad \rho_B = \frac{1-2v}{\sqrt{v}}. \tag{2}$$

This uses closed distribution functions and includes both signs and all real thresholds. No restricted-parameter sharpening is invoked.

Lemma 2 (A smooth replacement with its complete remainder)**.** *Let $U_1, \dots, U_n$ be independent copies of a centered variable with variance s and finite third absolute moment m_3. For every real t,*

$$\left|\mathbb{E}\Phi\left(\frac{t - np - \sum_i U_i}{\sqrt{nv}}\right) - \Phi\left(\frac{t-np}{\sqrt{n(v+s)}}\right)\right| \le \frac{\phi(0)}{6\sqrt{n}\,v^{3/2}}\left(m_3 + \sqrt{\frac{8}{\pi}}\,s^{3/2}\right). \tag{3}$$

Proof. For $f(w) = \Phi((t - np - w)/\sqrt{nv})$,

$$\|f'''\|_\infty \le \frac{\phi(0)}{(nv)^{3/2}}.$$

Indeed $\Phi'''(x) = (x^2 - 1)\phi(x)$ has absolute maximum $\phi(0)$: the other possible nonzero extrema are at $x = \pm\sqrt{3}$, with value $2e^{-3/2}\phi(0) < \phi(0)$.

Replace $U_1, \dots, U_n$ one at a time by independent $G_i \sim N(0, s)$. Conditional on all the other terms, Taylor's formula through degree two has remainder bounded by $\|f'''\|_\infty |x|^3/6$. Means and second moments cancel. Each replacement therefore costs at most $\|f'''\|_\infty (m_3 + \mathbb{E}|G_i|^3)/6$. The Gaussian third absolute moment is $\mathbb{E}|G_i|^3 = \sqrt{8/\pi}\, s^{3/2}$. Summing the n remainders gives the stated bound, since a Gaussian sum of variance ns convolved with the Gaussian CDF of variance nv gives variance $n(v + s)$. The argument is finite, uniform in t, and remains valid at $s = 0$. □

Lemma 3 (The moment increase pays for smoothing)**.** *Under Theorem 1's hypotheses,*

$$\rho - \rho_B \ge \frac{27}{400}\frac{s}{v^{3/2}}, \qquad \frac{\phi(0)}{6}\left(m_3 + \sqrt{\frac{8}{\pi}}s^{3/2}\right) \le \frac{3}{125}s. \tag{4}$$

Proof. Put $m_3 = \mathbb{E}|U|^3$ and $\mu_3 = \mathbb{E}U^3$. Write the containing interval as $[-a, b]$, with $a, b \ge 0$ and $d = a + b \le 1/5$; centering permits this choice. Then

$$|U| \le d, \qquad s \le ab \le d^2/4 \le 1/100, \qquad m_3 \le ds, \qquad |\mu_3| \le ds.$$

The variance bound follows by averaging $(U + a)(b - U) \ge 0$.

Since $|U| \le 1/5 < \min(p, 1 - p)$, the sign of $B - p + U$ is fixed by B. Expanding the two cubics gives the exact identity

$$\mathbb{E}|B - p + U|^3 = v(1 - 2v) + 6vs + (2p - 1)\mu_3. \tag{5}$$

For $x \ge 0$, Taylor's formula gives $(1 + x)^{3/2} \le 1 + \frac{3}{2}x + \frac{3}{8}x^2$, because the third derivative is negative. Consequently

$$\begin{aligned}(v + s)^{3/2}(\rho - \rho_B) &\ge s\left[9v - \frac{3}{2} - |2p - 1|d - \frac{3}{8}(1 - 2v)\frac{s}{v}\right] \\ &\ge s\left[\frac{3}{16} - \frac{1}{10} - \frac{1}{80}\right] = \frac{3}{40}s.\end{aligned}$$

Here $v \ge 3/16$, $|2p - 1| \le 1/2$, $1 - 2v \le 5/8$ and $s/v \le 4/75$. Also

$$\left(\frac{v}{v + s}\right)^{3/2} \ge \left(\frac{75}{79}\right)^{3/2} > \frac{9}{10}.$$

This proves the first bound in (4).

Finally $\phi(0) < 2/5$, $\sqrt{8/\pi} < 8/5$, and $\sqrt{s} \le 1/10$ give

$$\frac{\phi(0)}{6}\left(m_3 + \sqrt{\frac{8}{\pi}}s^{3/2}\right) \le \frac{1}{15}\left(\frac{1}{5} + \frac{8}{5}\frac{1}{10}\right)s = \frac{3}{125}s.$$

All constants are independent of n, the threshold and the law. □

2 Completion and scope

Proof of Theorem 1. Let $Y_i = B_i + U_i$, with all Bernoulli and noise variables independent. For every raw threshold t, condition on $W = \sum_i U_i$ and use (2). Lemma 2 then gives

$$\sqrt{n}\left|\mathbb{P}\Big(\sum_i Y_i \le t\Big) - \Phi\left(\frac{t-np}{\sqrt{n(v+s)}}\right)\right| \le C_{\mathrm{E}}\rho_B + \frac{3}{125}\frac{s}{v^{3/2}}.$$

The elementary inequality $C_{\mathrm{E}} > 2/5$ and Lemma 3 imply

$$C_{\mathrm{E}}(\rho - \rho_B) \ge \frac{27}{1000}\frac{s}{v^{3/2}}.$$

Since $27/1000 - 3/125 = 3/1000$, subtraction proves (1) after setting $t = np + H\sqrt{v+s}$ and taking the supremum. Conditioning retains every threshold atom; only the smooth Gaussian term was differentiated. □

The constant estimates need only $25/8 < \pi < 22/7$. The upper bound follows from

$$22/7 - \pi = \int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx > 0;$$

the integral of the alternating geometric polynomial through $-x^{126}$ in $(1+x^2)^{-1}$ gives the lower bound. Moreover $C_{\mathrm{E}}^2 > 37/(72\pi) > 259/1584 > 4/25$, since $\sqrt{10} > 3$. The companion rational checker verifies these finite inequalities and the exact cubic expansions on actual finite laws. It does not reprove Schulz's theorem.

If U_0 and U_1 are the two conditional noise laws in the acquired Bernoulli coordinates and their laws coincide, then B and U are independent. The common support lies in $[-a_1, a_0]$, so its width is at most .072 on the right and .068 on the left. The theorem therefore excludes every such acquired contrary witness, including positive noise above the currently certified regional ceilings. It also applies well outside that acquired setting.

This removes the count and Poisson comparison computations for precisely this common-noise subclass. It does not show that every extremizer has equal conditional noise laws, nor does it prove the finite base or the global conjecture. It bounds the sharp Berry–Esseen excess; it does not assert domination by the discrepancy at every chosen Bernoulli anchor. The unweighted crossing estimate and its phase counterexamples in `loss-common.tex` are supplementary and are not dependencies of this proof.

References

[1] J. Schulz, *The optimal Berry–Esseen constant in the binomial case*, doctoral dissertation, Universität Trier, 2016. Theorem 1, printed p. 1 (PDF page 11). `https://d-nb.info/1197702695/34`.